

PaySphere - API Contract Design

Version: 1.0

Purpose: Define all REST API contracts before implementation following industry best practices.

1. API Design Principles

Principle	Description
Versioning	All endpoints use /api/v1/
Resource Naming	Use nouns (e.g. /wallets) instead of verbs.
HTTP Verbs	GET, POST, PUT, DELETE determine actions.
Standard Response	Every API returns ApiResponse<T>.
Authentication	JWT Bearer Token for protected APIs.

2. Standard Response Wrapper

```
{
  "success": true,
  "message": "Operation completed successfully.",
  "data": {},
  "errors": [] }
```

3. Authentication Service

Register

Method	POST
Endpoint	/api/v1/auth/register
Authorization	No
Purpose	Register a new user
Request DTO	{"fullName","email","phoneNumber","password"}
Response DTO	ApiResponse<RegisterResponse>
Status Codes	201,400,409
Validation	FullName required; Email valid & unique; Phone unique; Password >=8 chars
Exceptions	DuplicateUserException

Login

Method	POST
Endpoint	/api/v1/auth/login
Authorization	No
Purpose	Authenticate user and issue JWT
Request DTO	{"email","password"}
Response DTO	ApiResponse<LoginResponse {token,expiresAt}>
Status Codes	200,401
Exceptions	InvalidCredentialException

Get Profile

Method	GET
Endpoint	/api/v1/auth/profile
Authorization	JWT
Purpose	Return logged-in user's profile
Request DTO	-
Response DTO	ApiResponse<ProfileResponse>
Status Codes	200,401

Change Password

Method	PUT
Endpoint	/api/v1/auth/change-password
Authorization	JWT
Purpose	Update password
Request DTO	{"currentPassword","newPassword"}
Response DTO	ApiResponse<object>
Status Codes	204,400,401

4. Wallet Service

Create Wallet

Method	POST
Endpoint	/api/v1/wallets
Authorization	JWT
Purpose	Create wallet for authenticated user
Request DTO	No Body
Response DTO	ApiResponse<CreateWalletResponse>
Status Codes	201,409
Exceptions	WalletAlreadyExistsException

Get Wallet

Method	GET
Endpoint	/api/v1/wallets/me
Authorization	JWT
Purpose	Get wallet details
Request DTO	-
Response DTO	ApiResponse<WalletResponse>
Status Codes	200,401

Top Up

Method	POST
Endpoint	/api/v1/wallets/top-up
Authorization	JWT
Purpose	Add money
Request DTO	{"amount"}
Response DTO	ApiResponse<WalletResponse>
Status Codes	200,400
Validation	Amount > 0

Withdraw

Method	POST
Endpoint	/api/v1/wallets/withdraw
Authorization	JWT
Purpose	Withdraw money
Request DTO	{"amount"}
Response DTO	ApiResponse<WalletResponse>
Status Codes	200,400,409
Validation	Amount > 0
Exceptions	InsufficientBalanceException

Transfer

Method	POST
Endpoint	/api/v1/wallets/transfer
Authorization	JWT
Purpose	Transfer funds
Request DTO	{"receiverEmail","amount","description"}
Response DTO	ApiResponse<TransactionResponse>
Status Codes	200,400,404,409
Validation	Receiver exists; Amount > 0; Cannot transfer to self
Exceptions	ReceiverNotFoundException, InsufficientBalanceException

Transaction History

Method	GET
Endpoint	/api/v1/wallets/transactions?page=1&pageSize=10&type=TopUp&sortBy=date&sortOrder=desc
Authorization	JWT
Purpose	Paginated transaction history
Request DTO	Query Parameters
Response DTO	ApiResponse<PagedTransactionResponse>
Status Codes	200

Get Transaction

Method	GET
Endpoint	/api/v1/wallets/transactions/{referenceNumber}
Authorization	JWT
Purpose	Retrieve transaction by reference
Request DTO	Path Parameter
Response DTO	ApiResponse<TransactionResponse>
Status Codes	200,404

5. Standard HTTP Status Codes

Code	Meaning
200	Success
201	Created
204	No Content
400	Validation Error
401	Unauthorized
403	Forbidden
404	Not Found
409	Business Conflict
500	Internal Server Error

6. Custom Exceptions

Authentication:

- DuplicateUserException
- InvalidCredentialException
- UserNotFoundException
- UnauthorizedAccessException

Wallet:

- WalletNotFoundException
- WalletAlreadyExistsException
- InsufficientBalanceException
- InvalidTransactionException
- ReceiverNotFoundException

7. DTO Organization

DTOs